

Kartikeya Vemula

Data Analyst | Product Analytics | Business Intelligence | Analytics Engineering
vnskartikeya@gmail.com | +1 (443) 630-1157 | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

Profile

Data Analyst with hands-on experience in SQL, Python, ETL pipelines, data warehousing, and business intelligence. Skilled in building executive dashboards, KPI monitoring systems, and data storytelling that translate raw data into decision-ready insights. Experienced across retail, product, operations, and engineering analytics, with strong stakeholder communication, cross-functional collaboration, and a process-improvement mindset.

Core Skills

Analytics: Data Analytics, Business Analysis, Business Intelligence, Product Analytics, Customer Analytics, KPI Monitoring, Performance Monitoring, Funnel Analysis, A/B Testing, Cohort Analysis, CLV, Retention Analytics, Churn Risk, Revenue Concentration, Market Basket Analysis, Root Cause Analysis

Data Engineering: ETL Pipelines, Data Warehousing, Data Modeling, Dimensional Modeling, Star Schema, Fact & Dimension Tables, Data Ingestion, PostgreSQL, BigQuery, REST APIs, Data Quality & QA

Tools & Visualization: Python, SQL, Excel, Power BI, Tableau, Plotly, Looker Studio, Dashboard Development, Executive Dashboards, Git, Docker, VS Code

Statistics & ML: Statistical Analysis, Predictive Analytics, Anomaly Detection (Rolling Z-Score), Artificial Intelligence (AI), Machine Learning, Forecasting, Time Series Analysis, XGBoost, SHAP

Professional: Problem Solving, Attention to Detail, Communication Skills, Verbal Communication, Stakeholder Communication, Stakeholder Management, Cross-Functional Collaboration, Executive Reporting, Decision Making, Process Improvement, Management

Projects

Customer Segmentation & Retention Intelligence Platform | *Python, SQL, PostgreSQL, RFM, CLV, Plotly*

- Built an RFM segmentation engine on 1M+ retail transactions, classifying 5,876 customers into 8 behavioral tiers using SQL window functions; Champions (22.1% of customers) drove 68.35% of total revenue, quantifying revenue concentration and retention ROI.
- Identified £171K+ revenue at risk across 15 high-value inactive customers; delivered cohort retention dashboards, CLV analysis, and executive reporting to support lifecycle marketing and stakeholder decision-making.

Instacart Customer & Operations Intelligence Platform | *Python, PostgreSQL, Docker, SQL, Apriori, Plotly Dash*

- Engineered a Dockerized PostgreSQL warehouse ingesting 3.4M+ orders across 206K customers and 49K SKUs for customer behavior, merchandising, and reorder analytics.
- Built RFM segmentation identifying 55,092 at-risk customers (26.7% of base); executed market basket analysis via Apriori (3.26 lift, 29.1% confidence) to surface basket affinity and product association rules.
- Developed a 4-tab Plotly Dash dashboard (Executive Overview, Customer Lifecycle, Merchandising, Market Basket) connected live to PostgreSQL, supporting cross-functional reporting on reorder rates and SKU velocity.

Customer Intelligence Data Warehouse | *Python, PostgreSQL, Docker, SQL, ETL, Star Schema, Power BI*

- Ingested, validated, and cleaned 1,067,371 raw e-commerce transactions under data quality and QA controls; built a star-schema model with fact and dimension tables for AOV, repeat purchase rate, and LTV across 43 countries.
- Developed executive dashboards in Power BI and Looker Studio for geographic BI, performance monitoring, and business reporting.

KPI Reliability & Diagnostic Engine | *Python, SQL, BigQuery-Compatible Schema, Plotly, YAML*

- Built an end-to-end KPI monitoring pipeline with 0–100 data health scores, volume/null checks, and rolling z-score-based anomaly detection with structured JSON diagnostics.
- Automated root cause analysis for KPI drops, identifying Mobile/APAC/Safari as the primary segment-level offender (–9.2% change), reducing manual analyst investigation time and turning incidents into dashboard-ready summaries.

Engineering Analytics Data Warehouse & Insight Pipeline | *Python, GitHub REST API, SQL, ETL, PostgreSQL, Grafana*

- Designed an ETL pipeline ingesting 5,000+ commits across 50+ repositories into a star schema with idempotent upserts, pagination handling, and rate-limit-aware API logic.
- Delivered SQL window-function analytics for cumulative commits, monthly velocity, and engineering productivity reporting.

Education

M.S. Data Science, University of Maryland Baltimore County, USA

2024–2026

GPA: 3.78/4.0 | Focus: Machine Learning, Statistical Modeling, Experimental Design, Time Series Analysis

B.Tech Computer Science, GITAM University, India

2020–2024

GPA: 8.89/10 | First-Class with Distinction